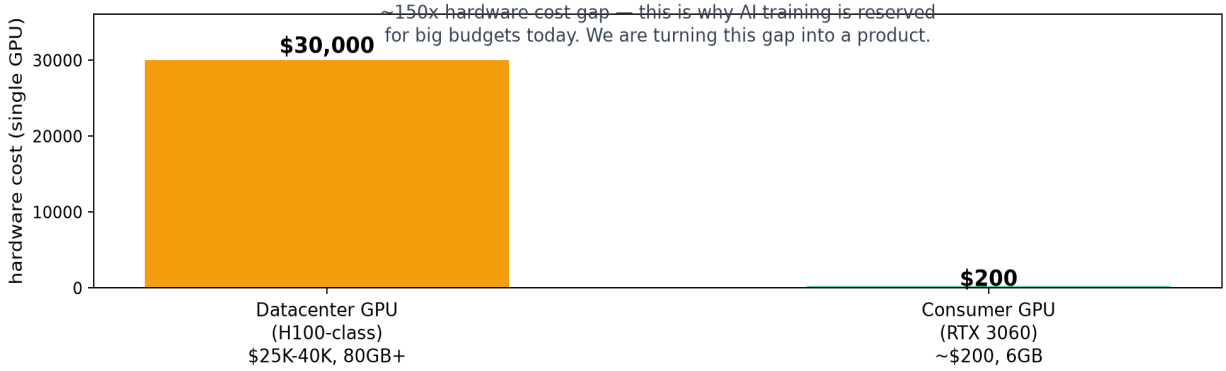
INVESTOR SUMMARY

We Fit Large Language Model Training on a 6GB Laptop GPU

Today, training a 1B+ parameter model requires $25-40K datacenter GPUs and six-figure training budgets. Our technology stack makes the same training runnable on a **~$200 consumer GPU (6GB)** — at a scale competitors cannot even attempt on the same hardware.



THE PROBLEM

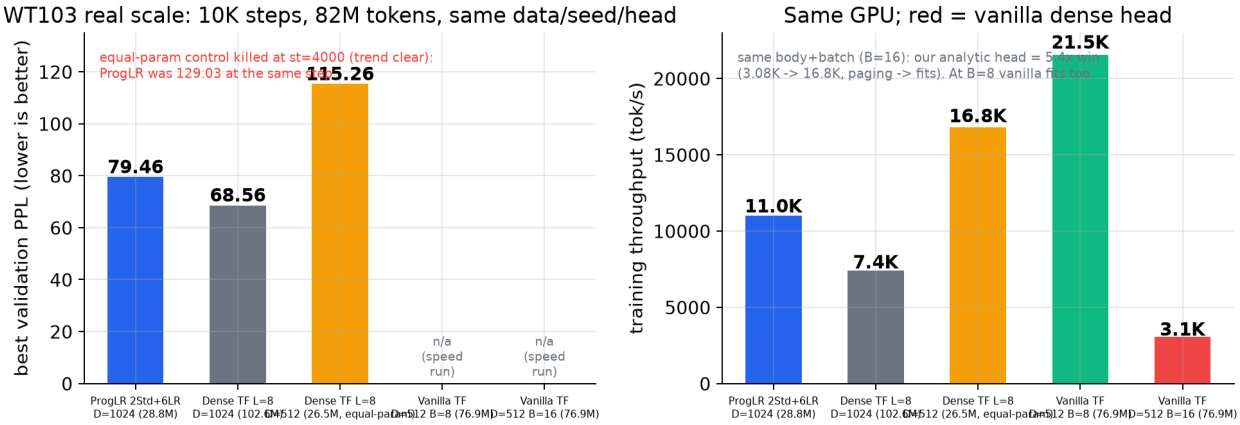
AI training has become a hardware race: as models grow, memory and compute needs compound and training costs run into the millions. The result: only big-budget players can participate. Meanwhile, millions of capable consumer GPUs (6-8GB) sit in the world, written off as "not enough" for AI training.

OUR PRODUCT — THREE TECHNOLOGIES, ONE GOAL: THE SAME JOB, MUCH CHEAPER

Technology	What it does	Measured gain
V3-Embed (vocabulary compression)	Compresses the model's vocabulary matrix into tiny learned factors — no vocabulary size	500x compression
Analytic Chunked Head (output-layer accelerator)	Computes training for 50K-250K+ class output layers with a memory-friendly method	5-12x speedup, enables very large vocabs
Progressive Low-Rank (slim hidden layers)	Converts part of the hidden layers to low-cost equivalents, fewer parameters, same model width	2-6x fewer parameters
NCE Sampled Head (throughput multiplier)	Trains the output layer on target + sampled negatives instead of the full vocabulary	2-4x extra speed, less compute

EVIDENCE — MEASURED ON THE SAME 6GB GPU

Measurement	Vanilla approach (competitor method)	Our technology
Training speed (same model, same batch)	3.1K tok/s — memory overflow (9.1GB)	16.8K tok/s — 5.4× faster, fits in 6GB
250K-vocabulary model	Cannot even start (output layer alone ~8GB)	Trained inside 6GB (~25 min final finetune)
WikiText-103 model (50K vocab)	Memory-paging at the same load	val PPL 95.18 — on 6GB, ~9 GPU-hours
229M-param dense model	Does not fit 6GB (constant paging)	Low-cost equivalent runs in 2.8GB
250K-vocab training speed (same model)	5.0K tok/s (exact CE)	14.9K tok/s — 3× with NCE
8.9M-param model training speed (same model)	20.6K tok/s	90.9K tok/s — 4.4×
Quality + speed (hybrid NCE + exact refine)	val PPL 224.21 in 14 min (exact CE 2000 steps)	233.10 in same time — 96% of quality, 3.5× less time t



Same hardware, same model body, same batch size: red bar = vanilla (competitor) method; blue/orange bars = our technology. Red point: at the scale vanilla cannot even attempt, we train.

WHY NOW, WHY US

• Market opening: when training costs drop 30-100x, thousands of companies and research teams that say "AI training is too expensive" can enter — with their own data, on their own hardware.

• Defensible product: the memory-bound optimizations win exactly at the scale competitors cannot apply on the same hardware (250K vocab, wide models). Four patentable components: V3-Embed, analytic chunked head, progressive low-rank, NCE sampled head.

• Chain of evidence: every result was measured on a single 6GB GPU with controlled comparisons; the training accelerator's mathematical equivalence was verified bit-by-bit.

• Low entry cost: the entire stack was developed on ~$200 hardware — productization targets the same hardware class.

HONEST LIMITS

In the spirit of transparency: at equal parameter counts and small scale, a classic dense architecture can remain ahead on quality. The NCE speed layer is an estimator, not the exact distribution — at equal step counts its val PPL trails exact CE by ~2.8×; its ~4.4× speed lets the same wall-clock budget consume far more data/model, and it scales independently of vocabulary size. Exact CE remains the metric for quality. Our win is at

the scale competitors **cannot attempt** (wide model + large vocabulary + consumer GPU) — and that is exactly the region our product targets. We claim a measurable language-model metric (val PPL) with comparable values; we do not claim "magic" like equal quality at fewer parameters.

NEXT STEPS

• Demo: live training on the same laptop + text generation from the 250K-vocabulary model (~100 words/sec).

• Technical report: the 12-section document with all measurements, charts and validation steps (FINAL_REPORT.pdf).

• Meeting: 30 minutes for investment proposal, roadmap and go-to-market plan.

V3 Project — LLM Training on a 6GB GPU | Draft, pre-meeting | August 2026